

(commands). Here is the shell script that is specific to this utility:

```
#!/bin/sh
ETHTOOL=/usr/sbin/ethtool
test -x $ETHTOOL || exit 0

# Find settings with a given prefix and print them as they appeared in
# /etc/network/interfaces, only with the prefix removed.
gather_settings () {
    env | awk -F= '/"$1"/{
        sub(/^"/,"");
        gsub(/"/,"");
        print tolower($1), $2
    }'
}

# Gather together the mixed bag of settings applied with -s/-change
SETTINGS=""
$(F_LINK_SPEED+ speed $IF_LINK_SPEED)
$(F_LINK_DUPLEX+ duplex $IF_LINK_DUPLEX)
"

# WOL has an optional pass-key
set -- $IF_ETHERNET_WOL
SETTINGS="$SETTINGS{1.+ vol $1}{2.+ soppass $2}"

# Autonegotiation can be on/off or an advertising mask
case "$IF_ETHERNET_AUTONEG" in
    *) :
on/off) SETTINGS="$SETTINGS autoneg $IF_ETHERNET_AUTONEG" ;;
*) SETTINGS="$SETTINGS autoneg on $IF_ETHERNET_AUTONEG" ;;
esac

[ -z "$SETTINGS" ] || $ETHTOOL -change "$IFACE" $SETTINGS

SETTINGS="$gather_settings ETHERNET_PAUSE"
[ -z "$SETTINGS" ] || $ETHTOOL --pause "$IFACE" $SETTINGS

SETTINGS="$gather_settings HARDWARE_IRQ_COALESCE"
[ -z "$SETTINGS" ] || $ETHTOOL --coalesce "$IFACE" $SETTINGS

SETTINGS="$gather_settings HARDWARE_DMA_RING"
[ -z "$SETTINGS" ] || $ETHTOOL --ring "$IFACE" $SETTINGS

SETTINGS="$gather_settings OFFLOAD"
[ -z "$SETTINGS" ] || $ETHTOOL --offload "$IFACE" $SETTINGS
```

Now, let's turn our attention to some of the configuration files. The */etc/resolv.conf* is the one that handles the 'name resolver' (and has a single keyword per line). It may have a domain, search or nameserver entry in each line. Here is a typical sample:

```
# Generated by NetworkManager
nameserver <ip address>
```

```
nameserver <ip address>
```

The keyword 'domain' is used to specify a local domain name and 'search' is used to specify alternate domain names. 'Nameserver' is used to identify the IP address of a DNS to handle queries.

The Avahi daemon also uses this file. Avahi is essentially a Multicast DNS Service Discovery mechanism (more details towards the end of this column), which allows you to find services and hosts running on a local network without manual configuration.

It checks the */etc/resolv.conf* file entries for nameservers. Here is the relevant code:

```
dns_reachable() {
    $(grep -q nameserver /etc/resolv.conf) || return 1;

    # If there is no local nameserver and no we have no global ip addresses
    # then we can't reach any nameservers
    if ! $(grep -q "nameserver 127.0.0.1" /etc/resolv.conf); then
        # Get addresses of all running interfaces
        ADDRS=$(LC_ALL=C ifconfig | grep 'addr:');
        # Filter out all local addresses
        ADDRS=$(echo "$ADDRS" | egrep -v "\[127|Scope:Host|Scope:Link]");
        if [ -z "$ADDRS" ]; then
            return 1;
        fi
    fi
}

return 0
}
```

The */etc/host.conf* file is another file that you can use to configure and manage the resolver. A typical sample is given below:

```
order hosts,bind
multi on
```

The above lines ask the resolver to look for the */etc/hosts* (see the description below) file before performing a query using the nameserver. It also instructs the resolver to add all the addresses in the */etc/hosts* file. You may refer to the box titled 'Configuration' for more details.

We have seen that the */etc/hosts* file is where the resolver would be looking for addresses and names of local hosts. It means that if you are adding the name of the host here, then the resolver will not use the nameserver to get the address; rather it will use the corresponding IP address found in the */etc/hosts* file. Normally, you could see entries for the loopback interface and local host names. Here is a sample entry:

```
127.0.0.1 localhost
127.0.1.1 GNU-BOX
```